

In Search of Objects: 50 years of wondering

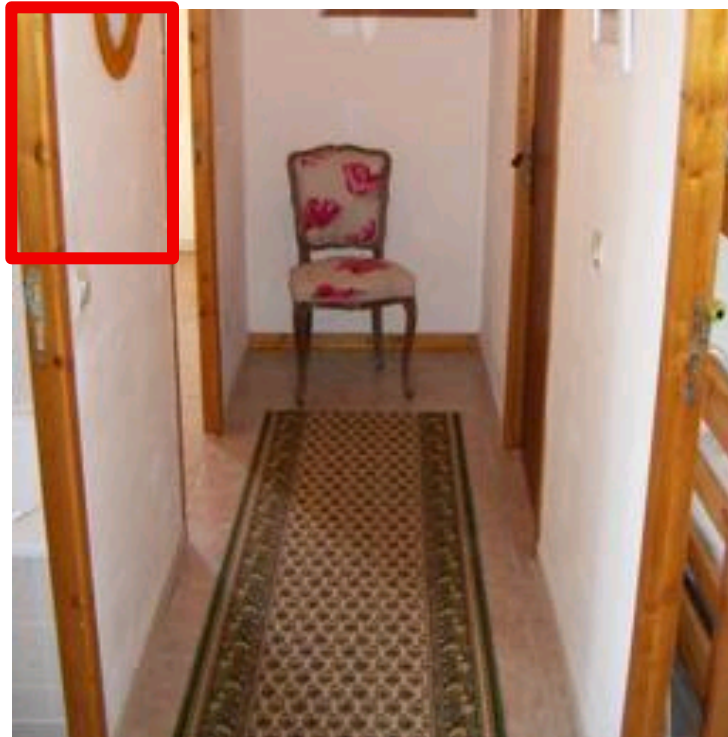


16-721: Learning-Based Methods in Vision
A. Efros, CMU, Spring 2009

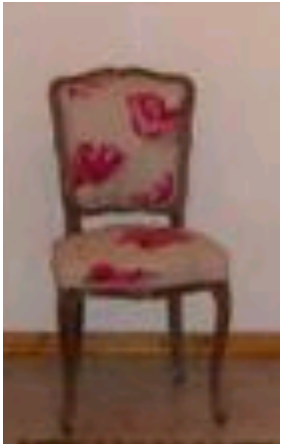
Object recognition

Is it really so hard?

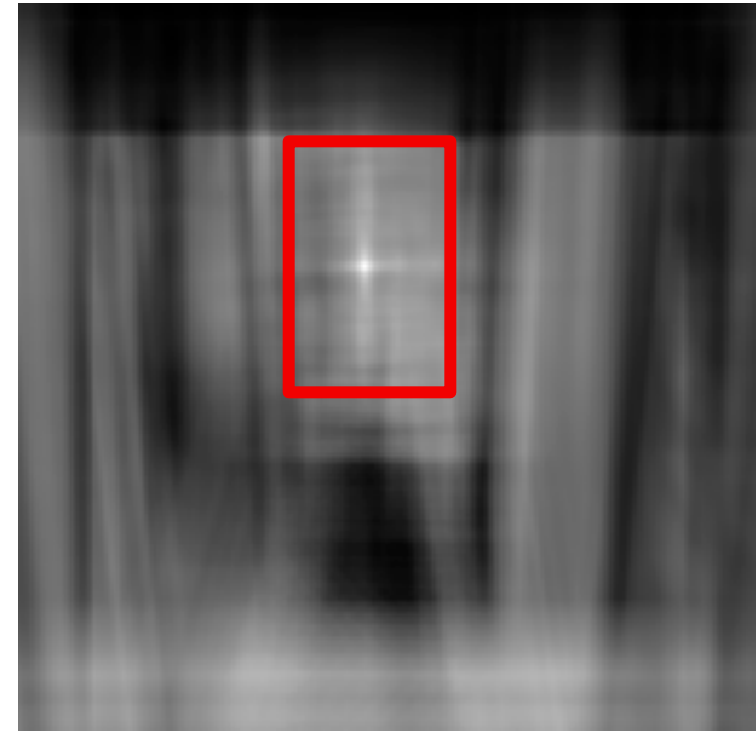
Find the chair in this image

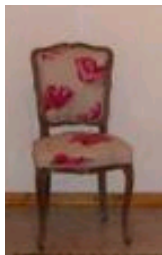


This is a chair



Output of normalized correlation

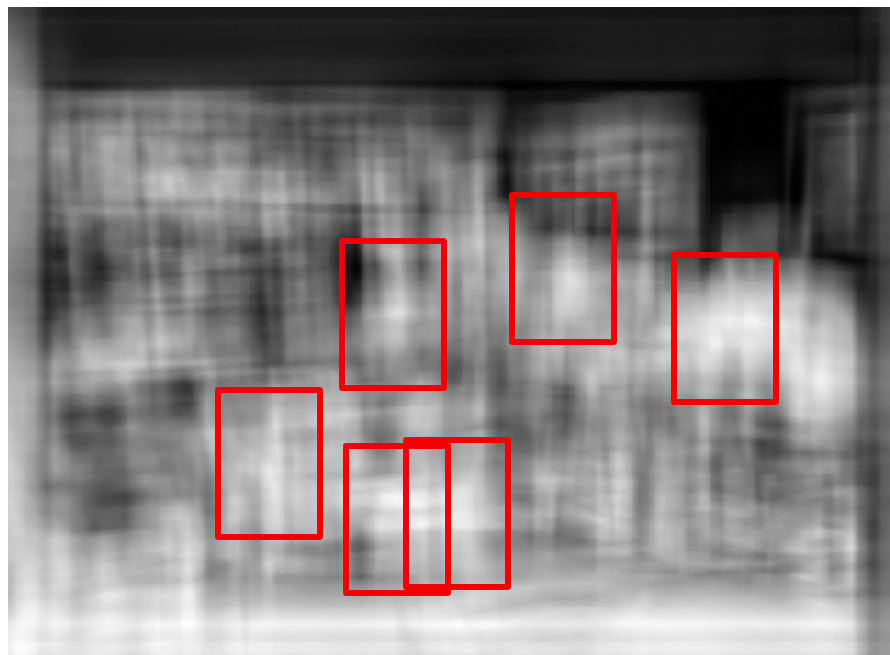
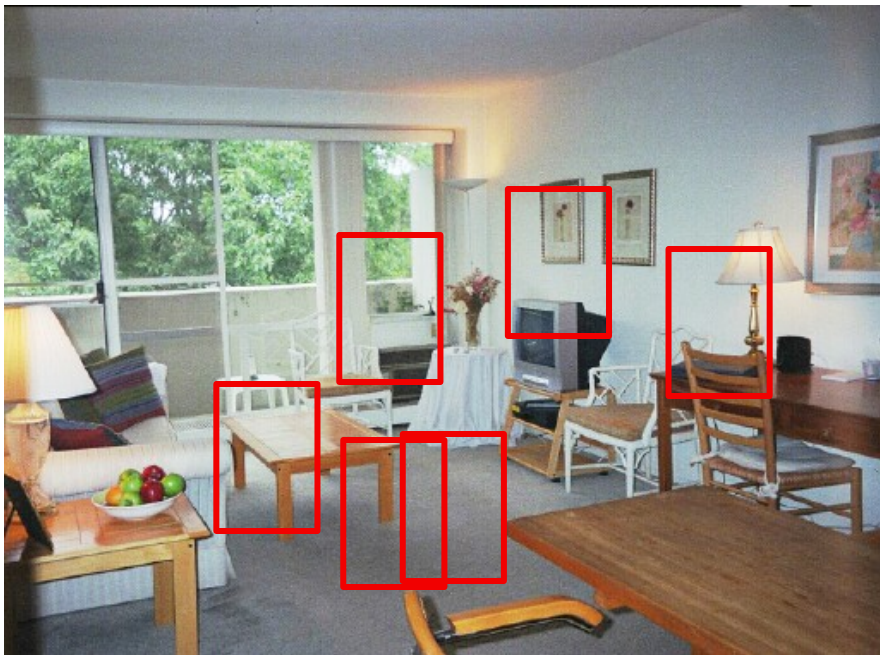




Object recognition

Is it really so hard?

Find the chair in this image



Pretty much garbage

Simple template matching is not going to make it

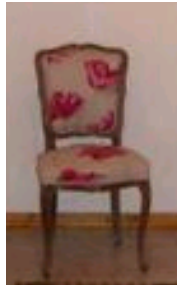
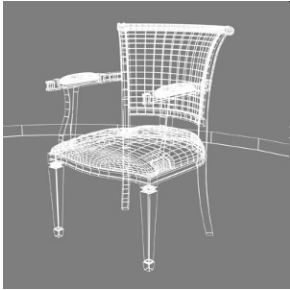
Antonio's biggest concern:

how do I justify 50 years of research if this experiment did work?

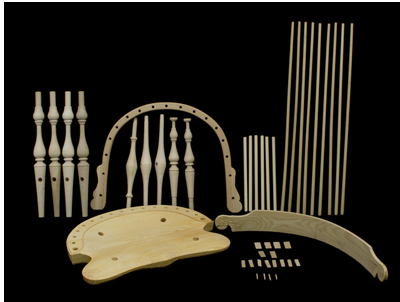
Slide by Antonio Torralba

The Religious Wars

Geometry vs. Appearance



Parts vs. The Whole



...and the standard answer:

probably both or neither

Geometry First

Roberts and the Blockworld (1960s)

If you don't like the world – get a new one!

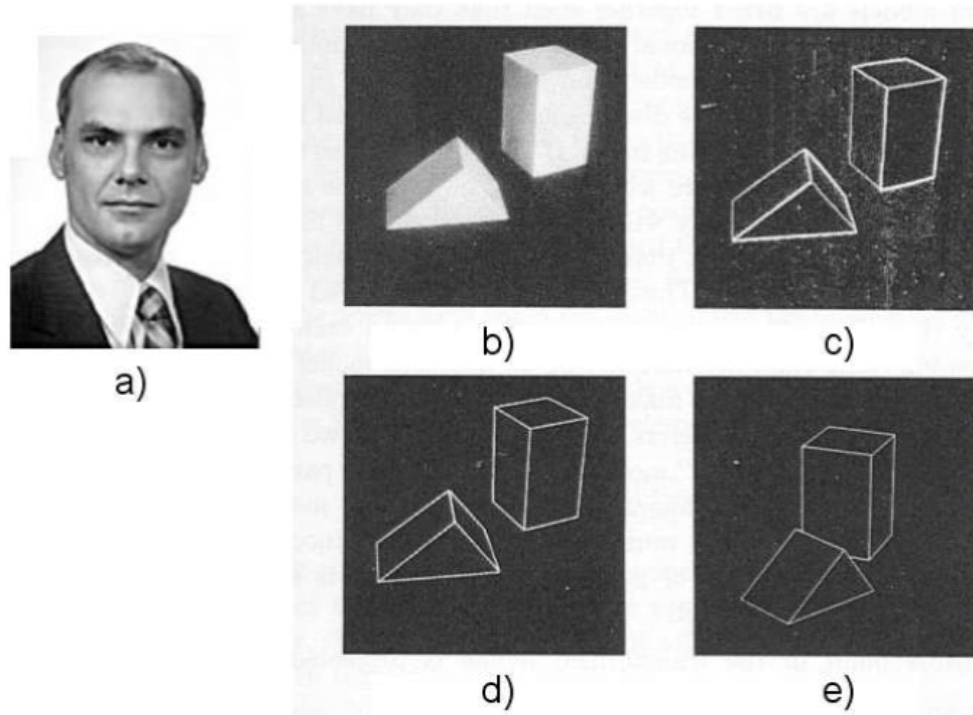


Fig. 1. A system for recognizing 3-d polyhedral scenes. a) L.G. Roberts. b) A blocks world scene. c) Detected edges using a 2x2 gradient operator. d) A 3-d polyhedral description of the scene, formed automatically from the single image. e) The 3-d scene displayed with a viewpoint different from the original image to demonstrate its accuracy and completeness. (b) - e) are taken from [64] with permission MIT Press.)

Binford and generalized cylinders (1970s)

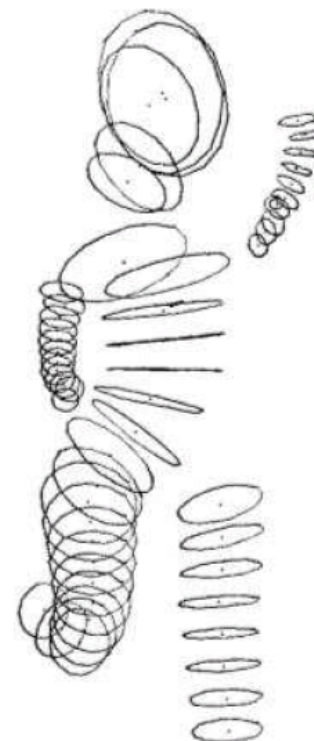
I am cylinder, you are a cylinder



a)



b)



c)

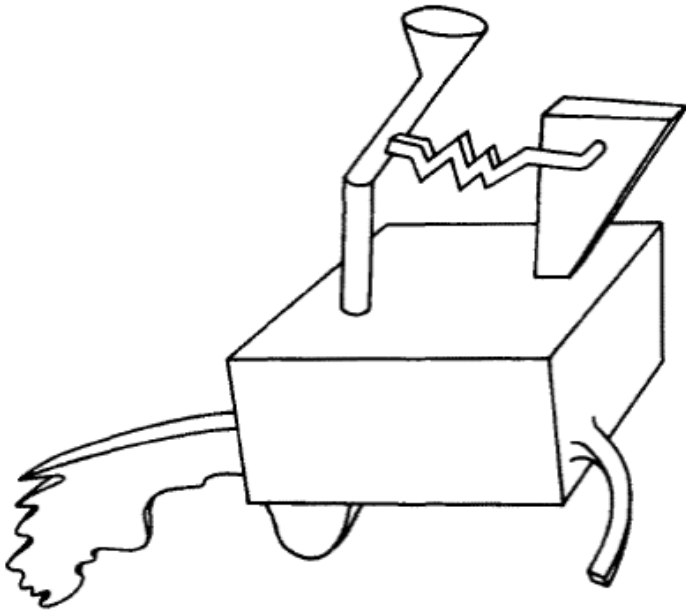
Fig. 3. The representation of objects by assemblies of generalized cylinders. a) Thomas Binford. b) A range image of a doll. c) The resulting set of generalized cylinders. (b) and c) are taken from Agin [1] with permission.)

Biederman and Recognition-by-components



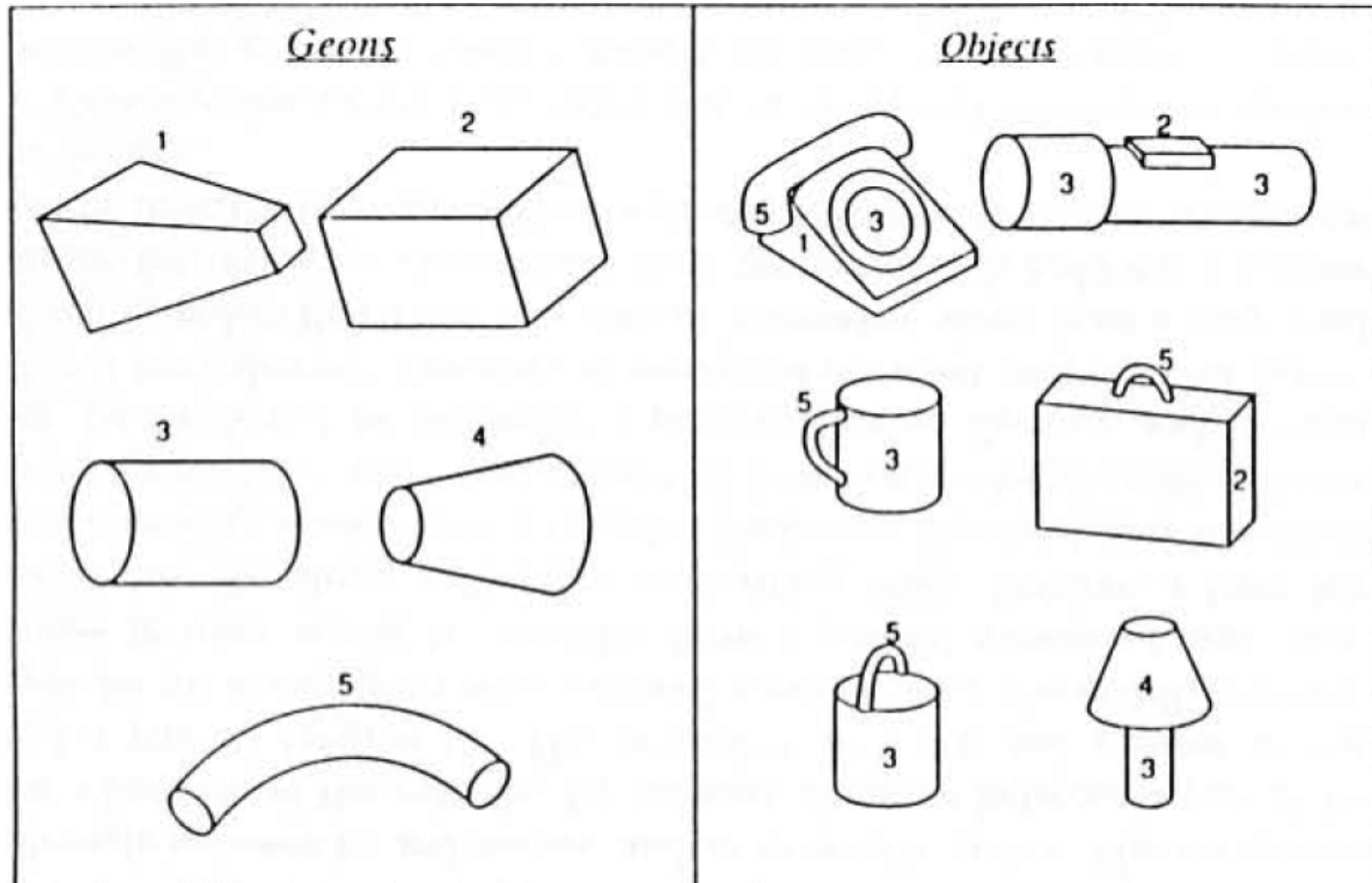
Irving Biederman

Recognition-by-Components: A Theory of Human Image Understanding.
Psychological Review, 1987.



- 1) We know that this object is nothing we know
- 2) We can split this objects into parts that everybody will agree
- 3) We can see how it resembles something familiar: “a hot dog cart”

Objects and their geons



Hypothesis: there is a small number of geometric components that constitute the primitive elements of the object recognition system (like letters to form words).

Aspect Graphs and their demise

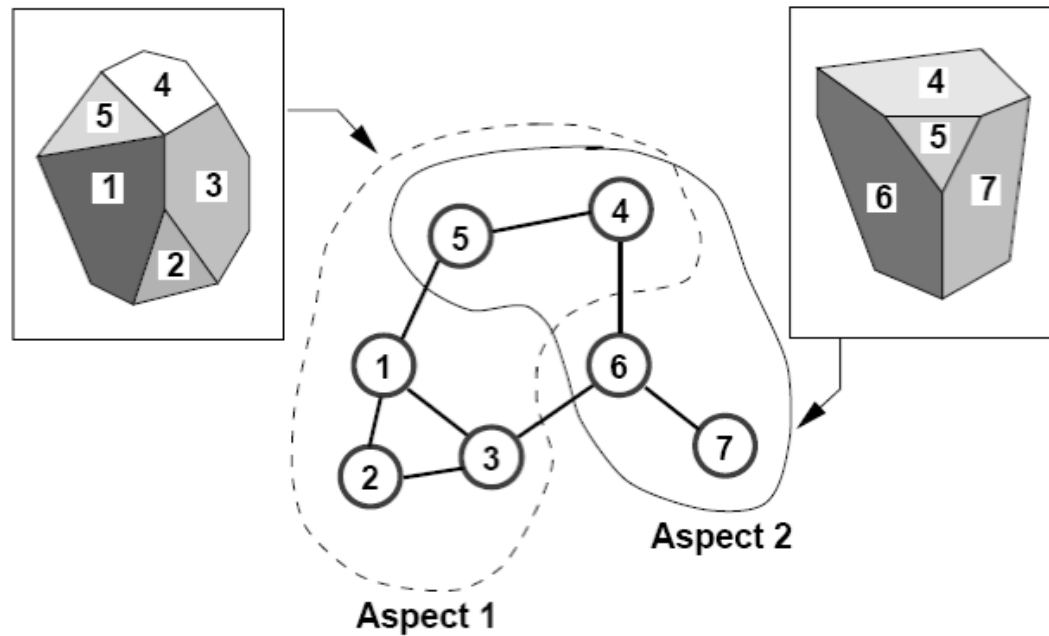
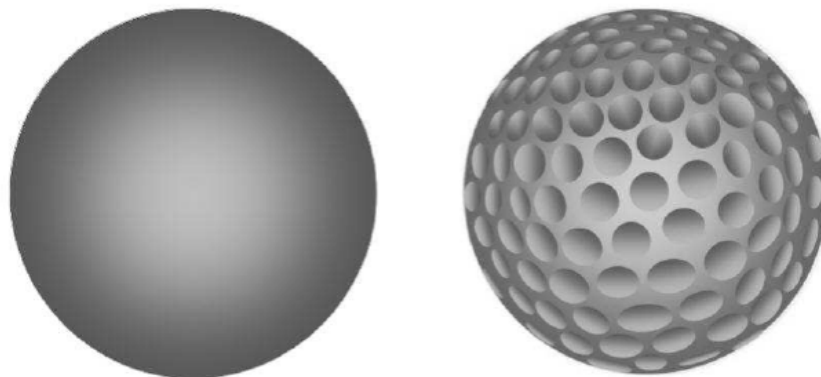


Fig. 6. Two views of a polyhedral solid. The adjacency of projected polygonal faces forms a graph. The view-based description is learned by associating new view structures with the existing graph. The figure is similar to one from [80].



Appearance Makes an Appearance

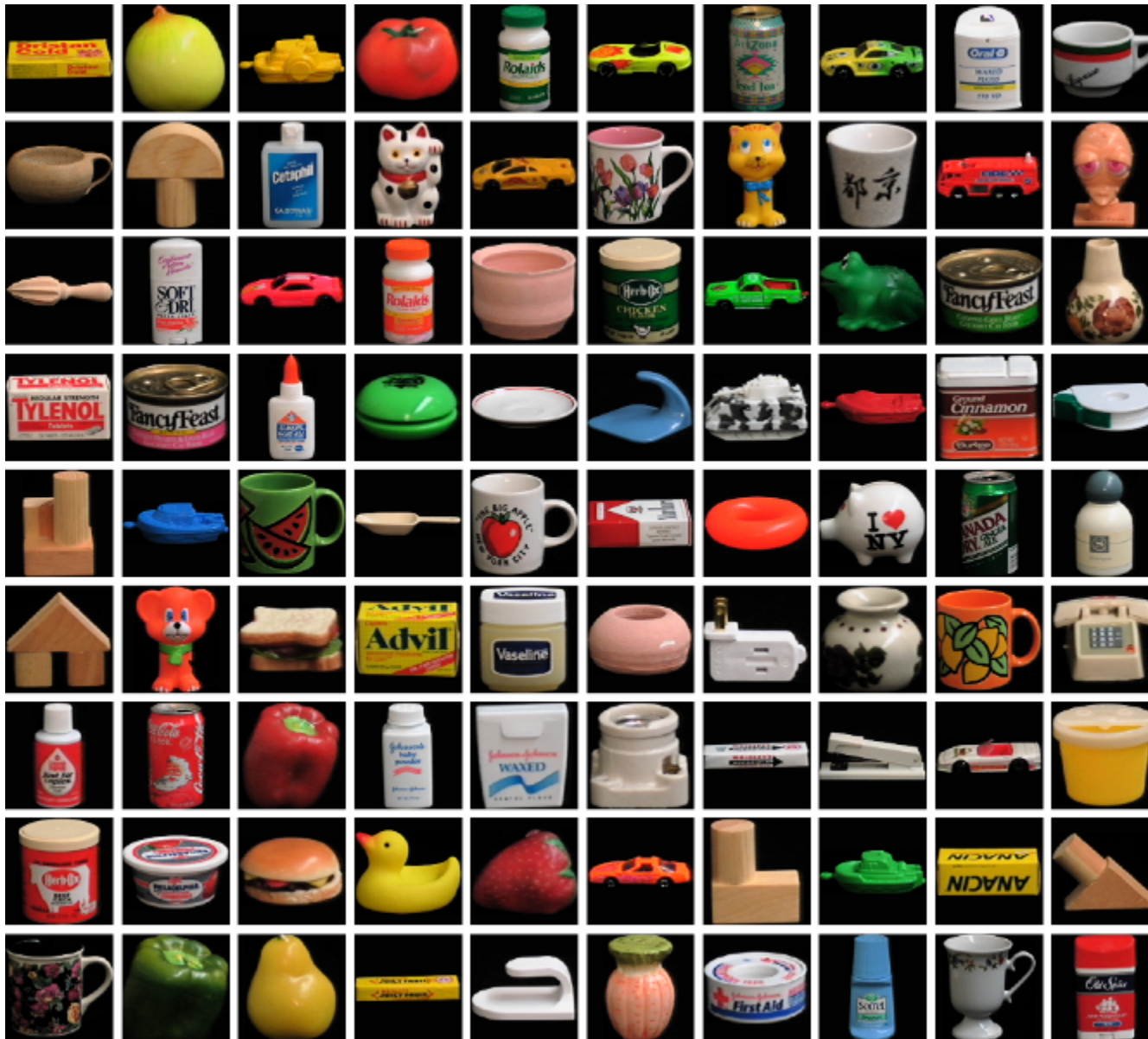
Eigenfaces: NN in low-dim subspace (1990s)



Later turns out,
simple NN works
Just as well...

Sirovich & Kirby (1987), Turk & Pentland (1991)

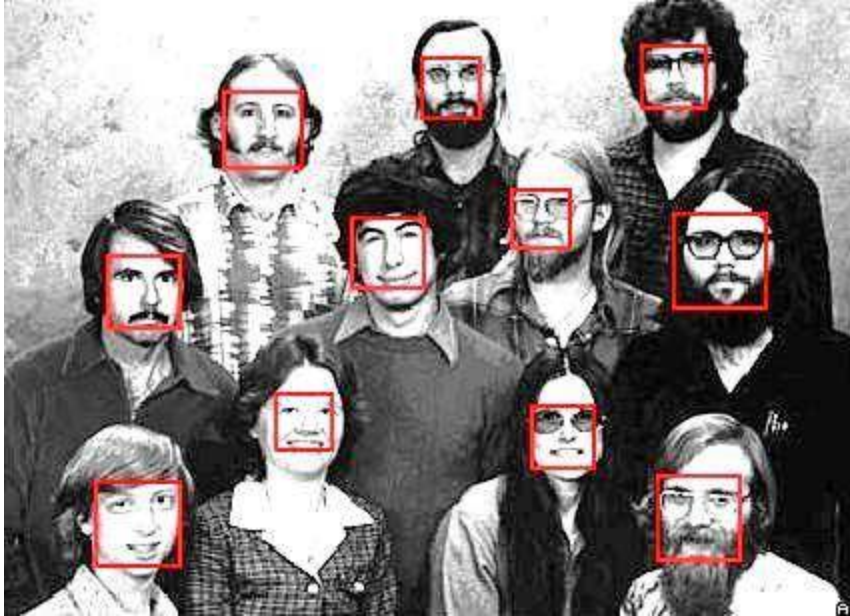
Columbia Object Image Library (COIL), 1996



Squash 3D pose variation with data!

Object not cropped? No problem!

The Age of Sliding Window Crazyiness

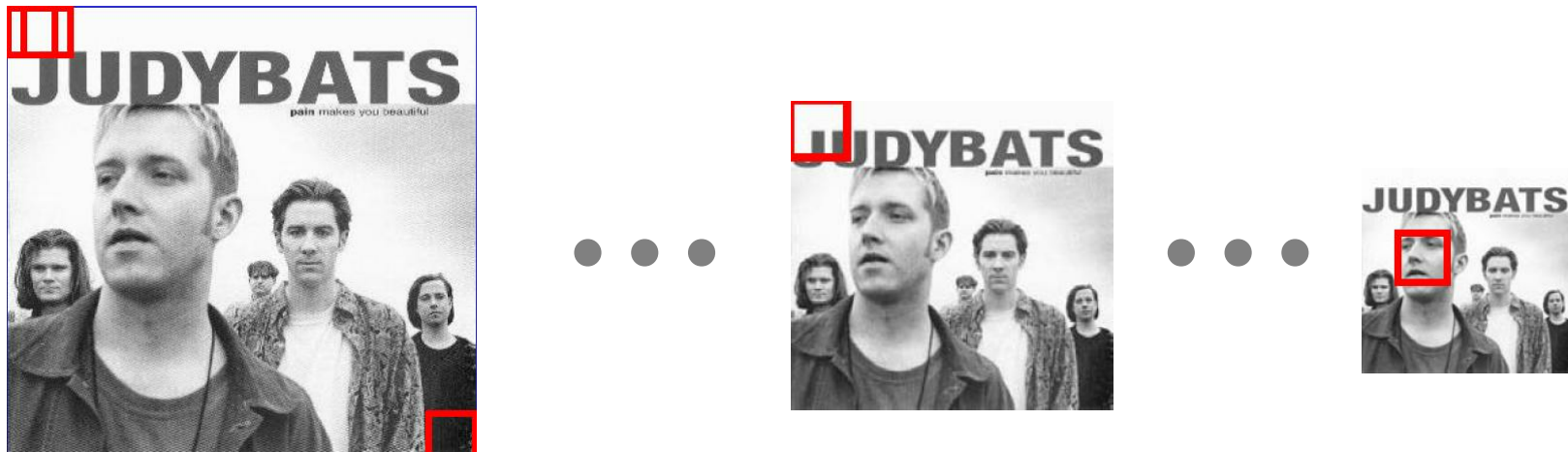


- Rowley et al., 1998
- Schniderman & Kanade, 1999
- Viola & Jones, 2001
- etc.



What *is* a Sliding Window Approach?

Search over space and scale



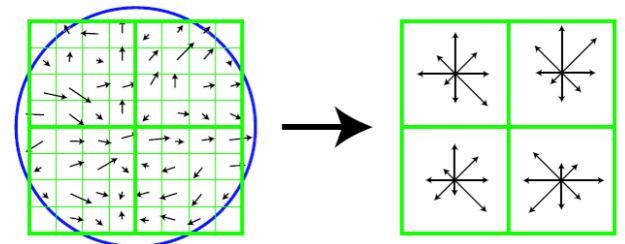
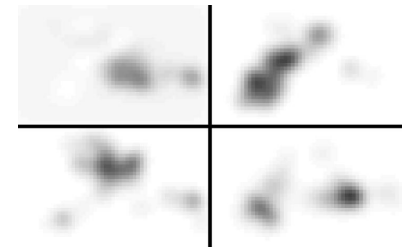
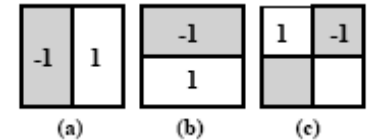
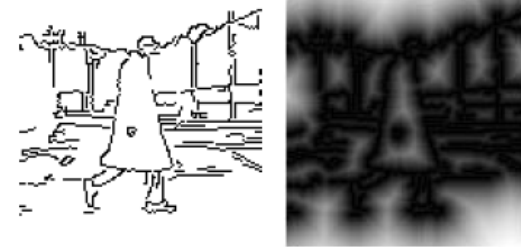
Detection as subwindow classification problem

“In the absence of a more intelligent strategy, any global image classification approach can be converted into a localization approach by using a sliding-window search.”

What features to match?

SSD is too strict. Need a bit of invariance to appearance, focus, and contours

- Edges (Chamfer/Housdorff/...)
- Wavelets / Filters / Jets ...
- Blur (Geometric Blur, ...)
- Spatial Histograms (SIFT, HOG, gist, Shape Context, ...)



Edge Matching



?

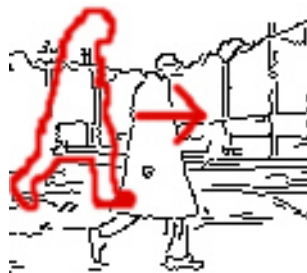


Edge-Template

(hand-drawn from footage, or automatically generated from CAD models)

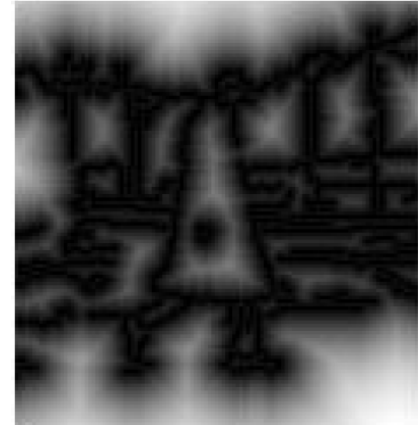
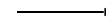
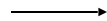
Image Scene

Real world, real time video footage.



Template sliding

Chamfer / Hausdorff Distance



Edge Map

Distance Transform

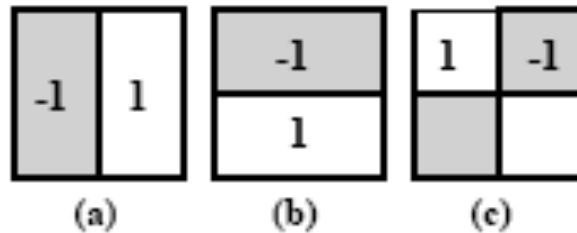
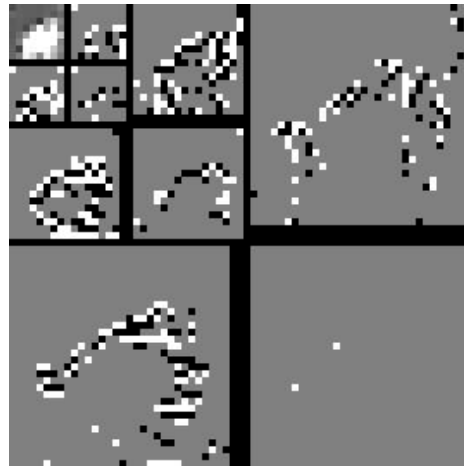
The Chamfer distance is the average distance to the nearest feature.

$$D_{chamfer}(T, I) \equiv \frac{1}{|T|} \sum_{t \in T} d_I(t)$$

Housdorff is distance of the worst matching object pixel to its closest image pixel.

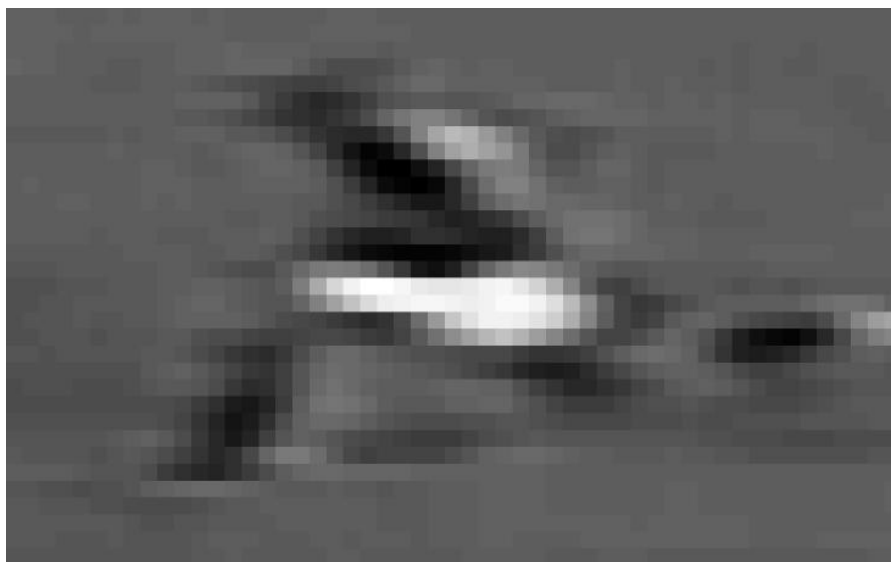
$$h(M, I) = \max_{m \in M} \min_{i \in I} \|m - i\|$$

Wavelets / Filters / Jets

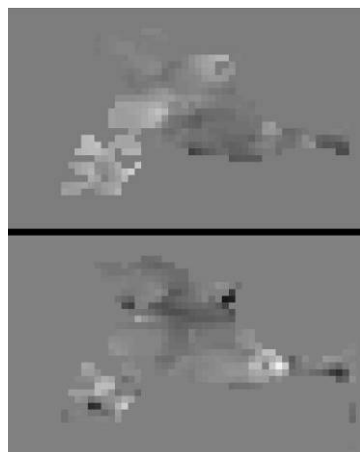


Schniderman & Kanade,
1999
Viola & Jones, 2001

blurring

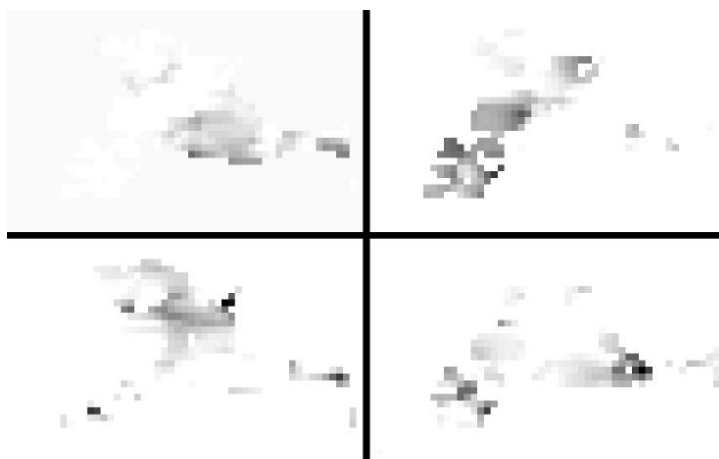


gradients



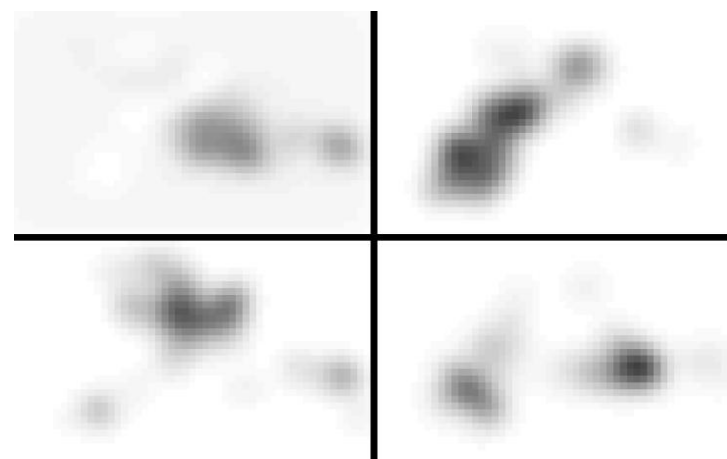
F_x, F_y

Half-wave rect.



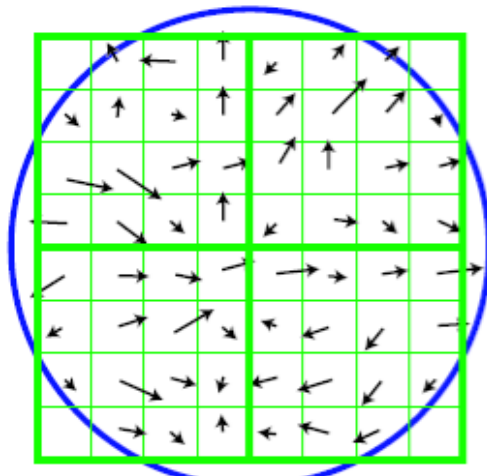
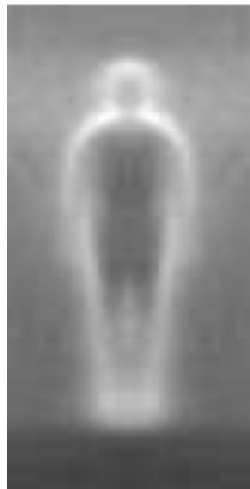
$F_x^-, F_x^+, F_y^-, F_y^+$

blur

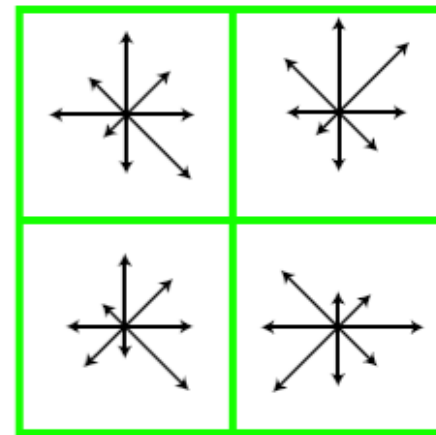


blurred $F_x^-, F_x^+, F_y^-, F_y^+$

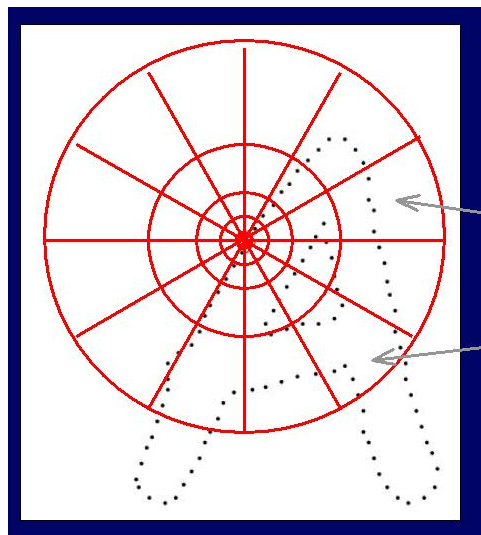
histograms (of gradients)



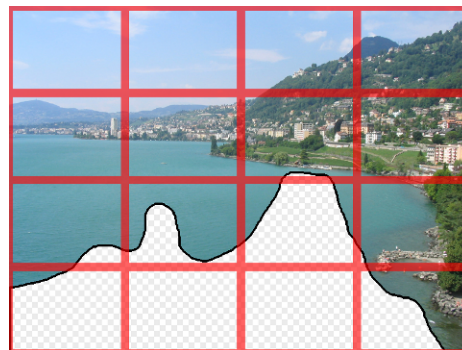
Gradients within 8X8 patch



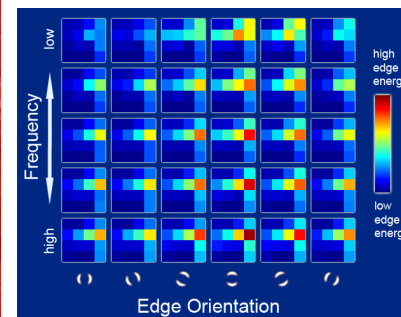
Bin into local (4X4) neighborhoods
& 8 orientations



Shape Context



Gist



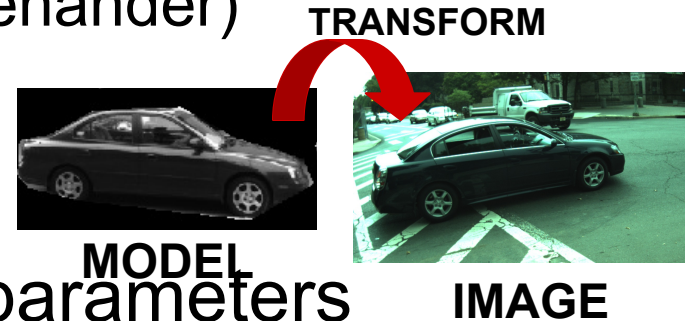
Freeman and Roth IAFGR 1995
Lowe ICCV1999
Oliva & Torralba, 2001
Belongie et al, 2001
Dalal & Triggs CVPR05

Binning achieves invariance to small patch offsets

Matching Parts

Why Matching?

- Old idea
 - Statistical Pattern Theory (Ulf Grenander)
 - Deformable Templates
 - Fischler & Elschlager
 - Etc. at least by the early 1970's
- “transform” and “appearance” parameters

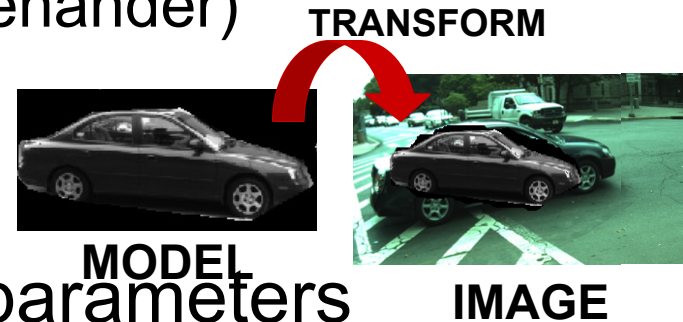


$$M(T, A)$$

- Matching to estimate transform

Why Matching?

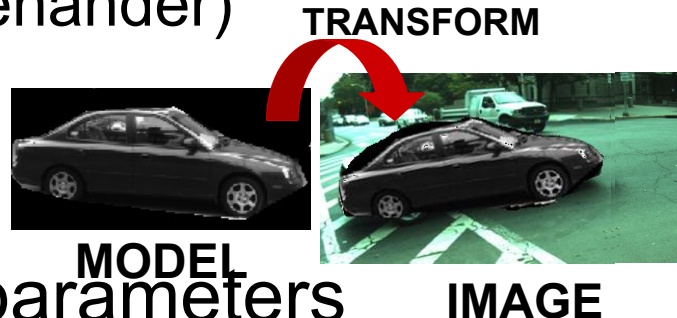
- Old idea
 - Statistical Pattern Theory (Ulf Grenander)
 - Deformable Templates
 - Fischler & Elschlager
 - Etc. at least by the early 1970's
- “transform” and “appearance” parameters



$$M(T, A)$$

- Matching to estimate transform

Why Matching?

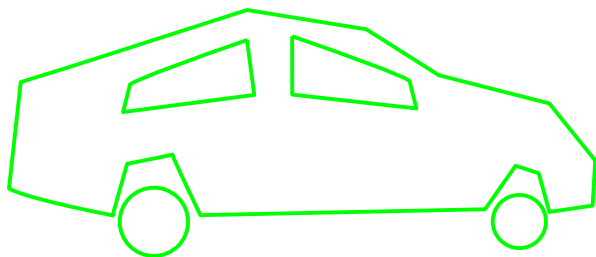
- Old idea
 - Statistical Pattern Theory (Ulf Grenander)
 - Deformable Templates
 - Fischler & Elschlager
 - Etc. at least by the early 1970's
 - “transform” and “appearance” parameters
- 
- The diagram illustrates the transformation process. On the left, a black car is shown against a black background, labeled 'MODEL'. On the right, the same car is shown in a real-world scene (a parking lot with a green truck in the background), labeled 'IMAGE'. A red curved arrow labeled 'TRANSFORM' points from the model to the image, indicating the mapping from the model to the image.

$$M(T, A)$$

- Matching to estimate transform
 - Searching over diffeomorphisms **difficult**
 - Searching over discrete assignments **easier?**

Why parts?

Model of Car



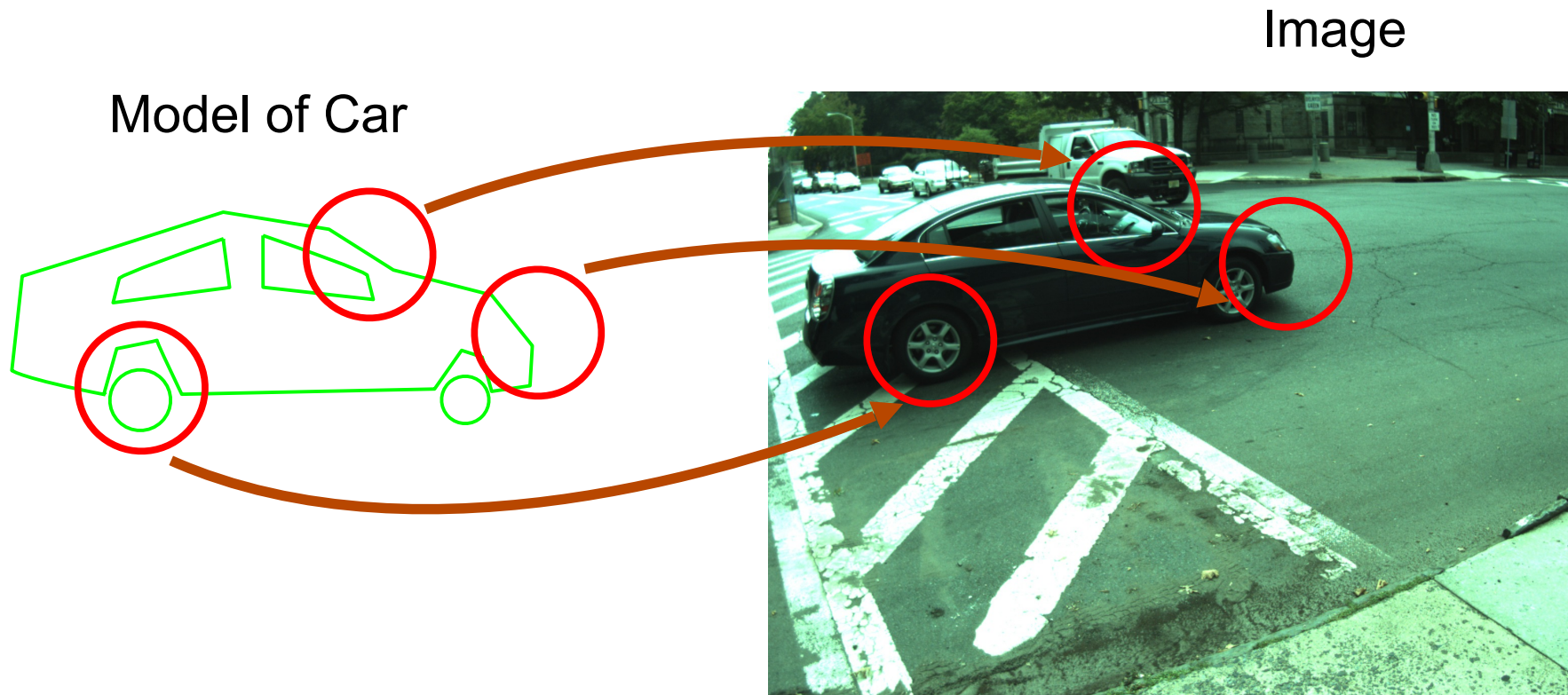
?



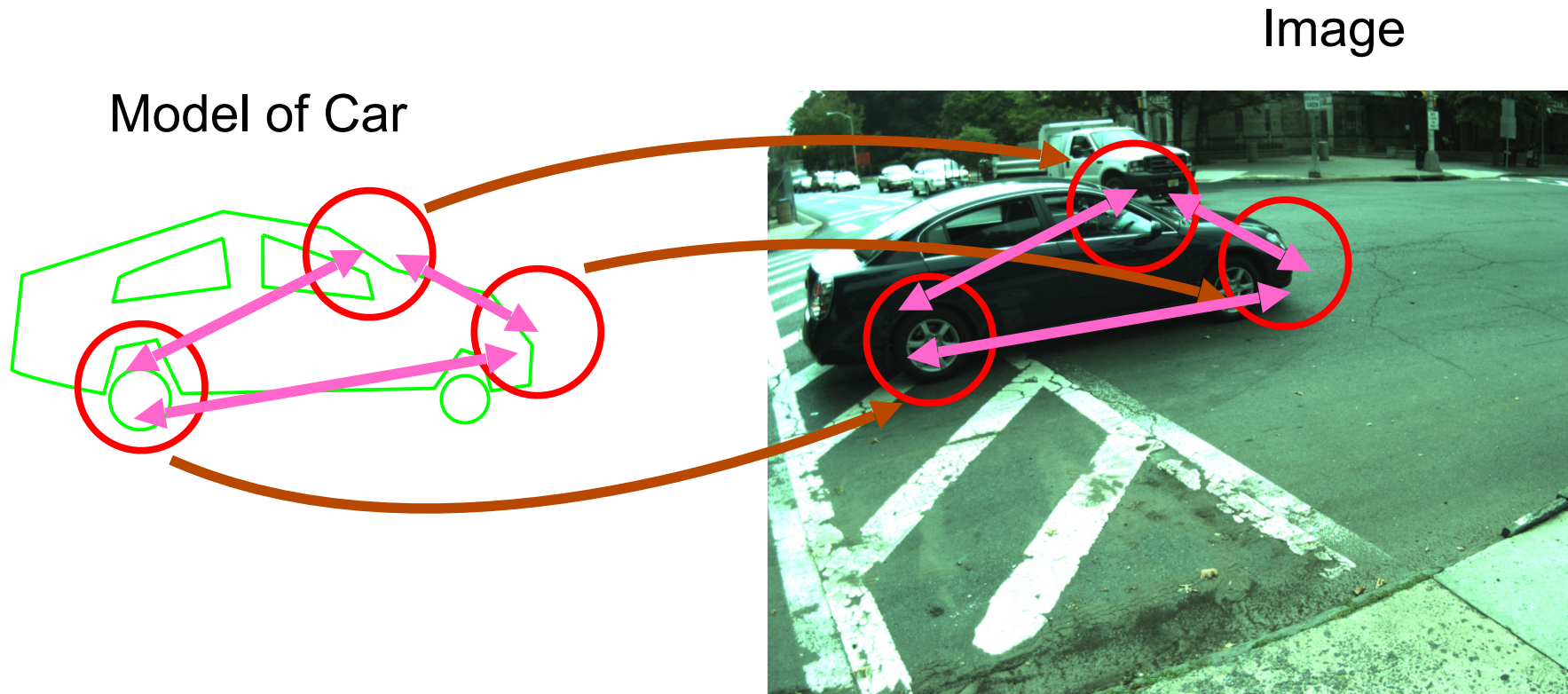
Image



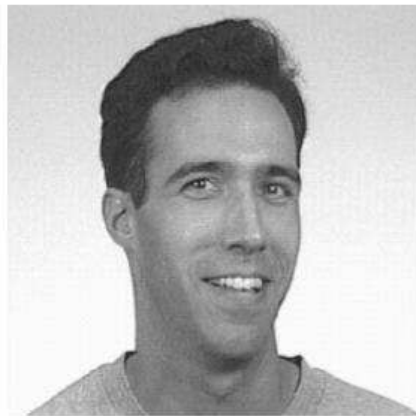
Why Parts?



Why Parts?



Huttenlocker & Ullman and Alignment



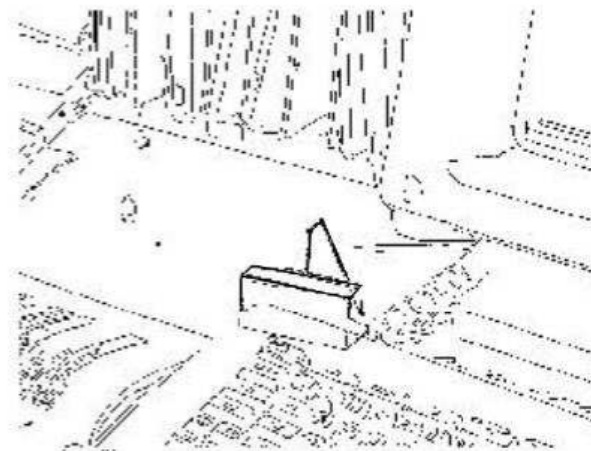
a)



c)



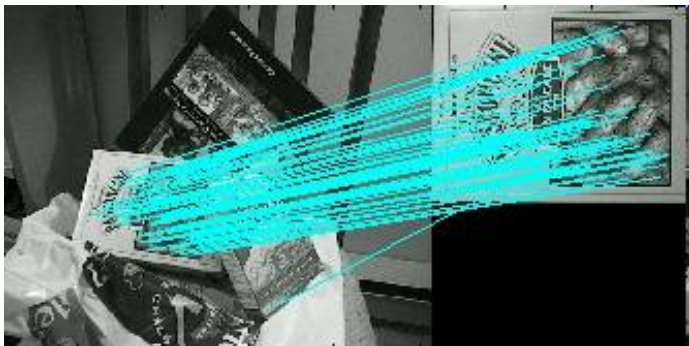
b)



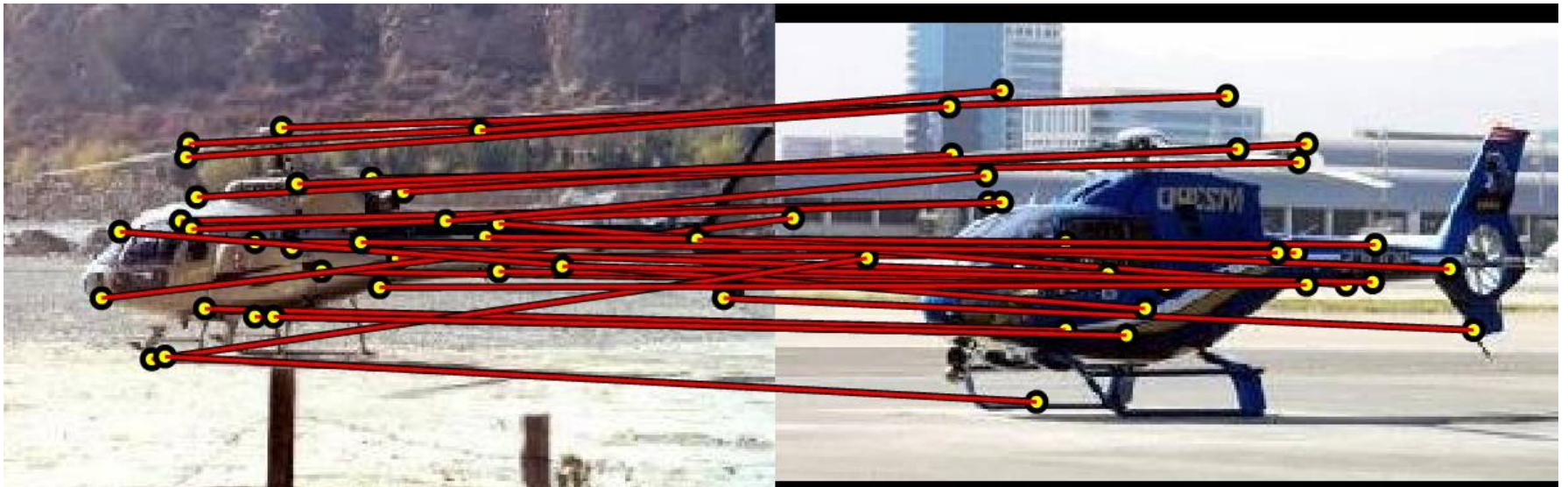
d)

Fig. 11. Three-dimensional object recognition using alignment. a) Dan Huttenlocher. b) Shimon Ullman. c) A cluttered image. d) The aligned model, shown near the middle of the image. (c) and d) provided by Dan Huttenlocher, with permission.)

Lowe and the birth of SIFT (1999)



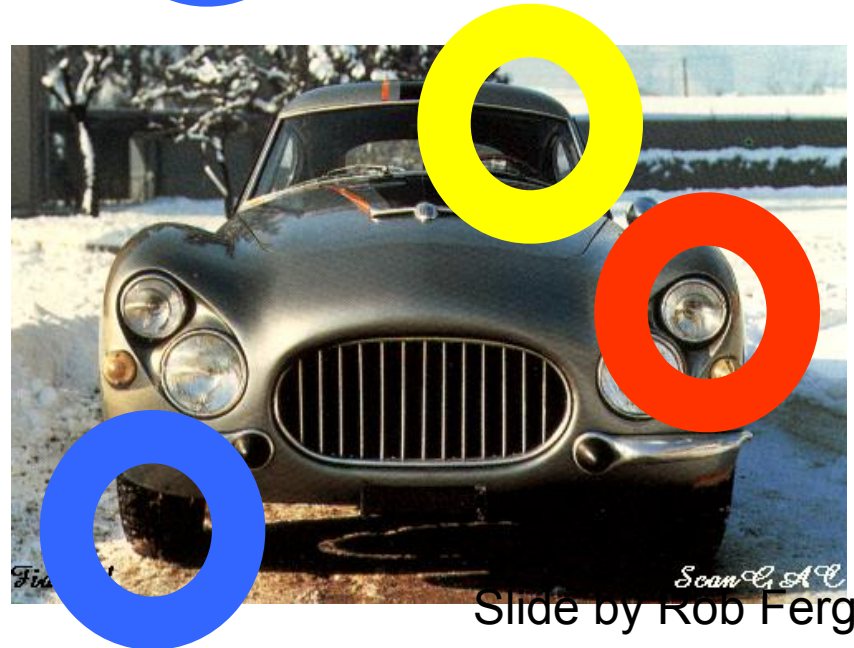
On to object classes!



Quadratic Assignment (Adding Geometric Constraints)

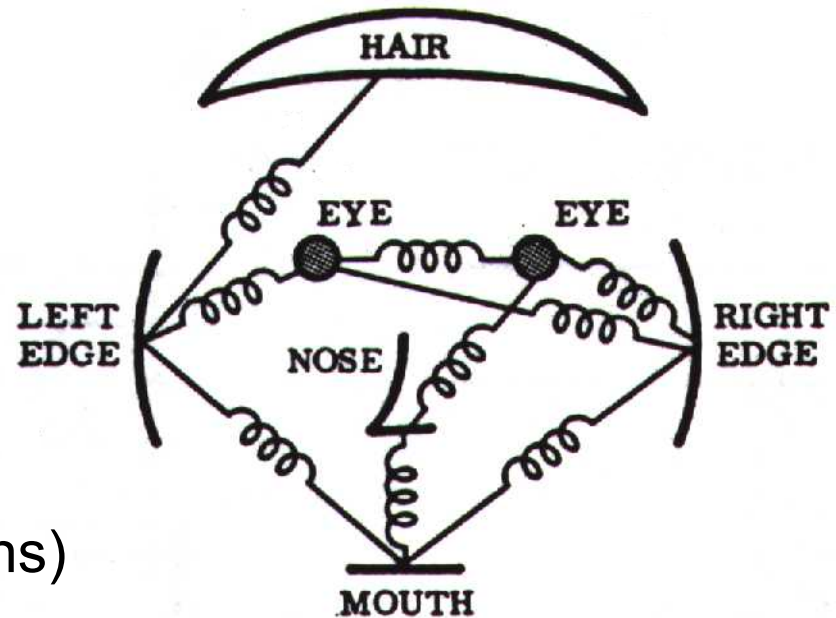


Model: Parts and Structure



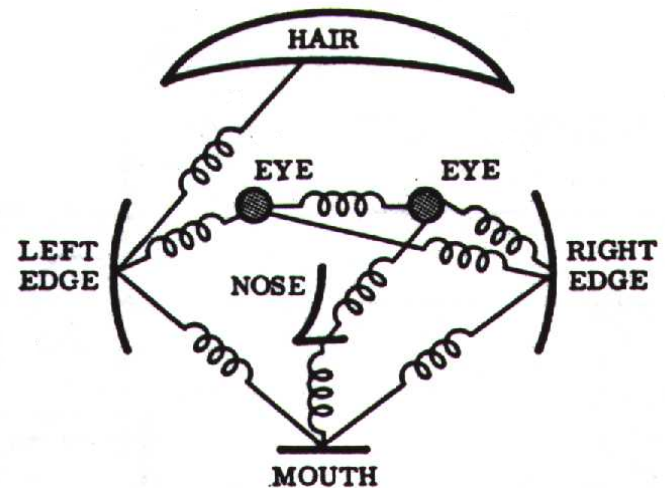
Representation

- Object as set of parts
 - Generative representation
- Model:
 - Relative locations between parts
 - Appearance of part
- Issues:
 - How to model location
 - How to represent appearance
 - Sparse or dense (pixels or regions)
 - How to handle occlusion/clutter



History of Parts and Structure approaches

- Fischler & Elschlager 1973
- Yuille '91
- Brunelli & Poggio '93
- Lades, v.d. Malsburg et al. '93
- Cootes, Lanitis, Taylor et al. '95
- Amit & Geman '95, '99
- Perona et al. '95, '96, '98, '00, '03, '04, '05
- Felzenszwalb & Huttenlocher '00, '04
- Crandall & Huttenlocher '05, '06
- Leibe & Schiele '03, '04
- Many papers since 2000



Constellation Models

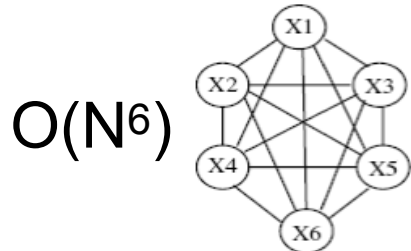
- + Sparse representation
- + Computationally tractable (10^5 pixels $\rightarrow$ 10^1 -- 10^2 parts)
- + Avoid modeling global variability



- Throw away most image information
- Parts need to be distinctive to separate from other classes

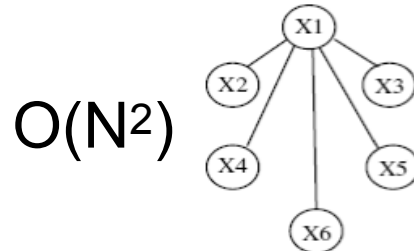
Different connectivity structures

Fergus et al. '03
Fei-Fei et al. '03



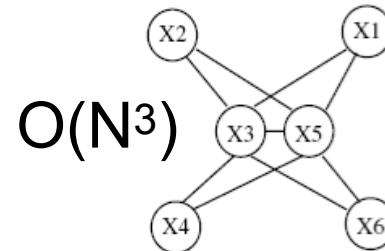
a) Constellation [13]

Crandall et al. '05
Fergus et al. '05



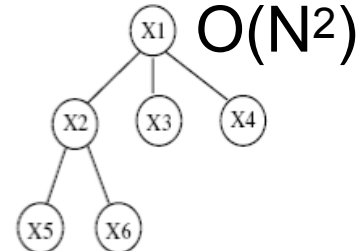
b) Star shape [9, 14]

Crandall et al. '05

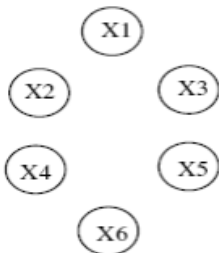


c) k -fan ($k = 2$) [9]

Felzenszwalb &
Huttenlocher '00

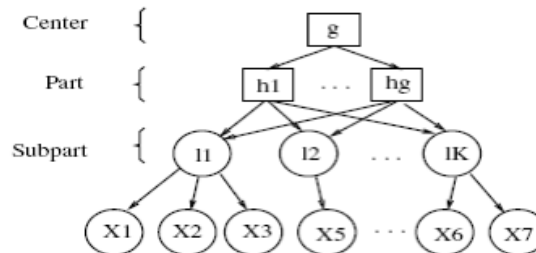


d) Tree [12]



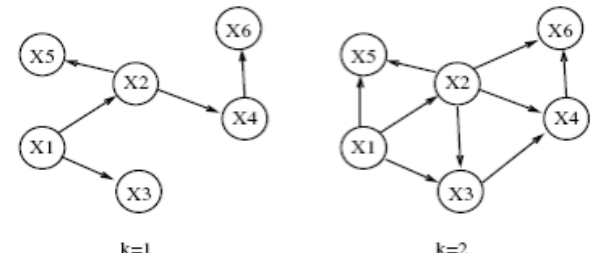
e) Bag of features [10, 21]

Csurka '04
Vasconcelos '00



f) Hierarchy [4]

Bouchard & Triggs '05



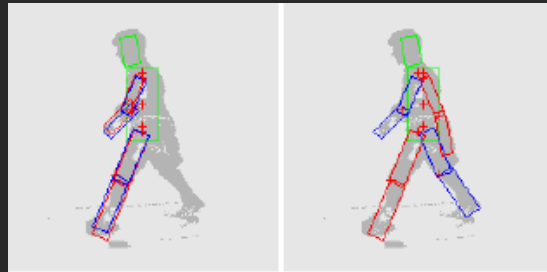
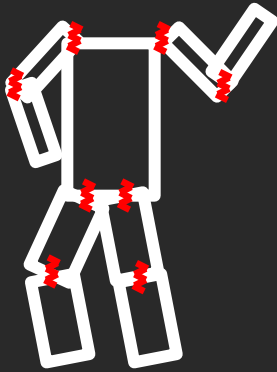
g) Sparse flexible model

Carneiro & Lowe '06

from Sparse Flexible Models of Local Features
Gustavo Carneiro and David Lowe, ECCV 2006

Trouble with trees

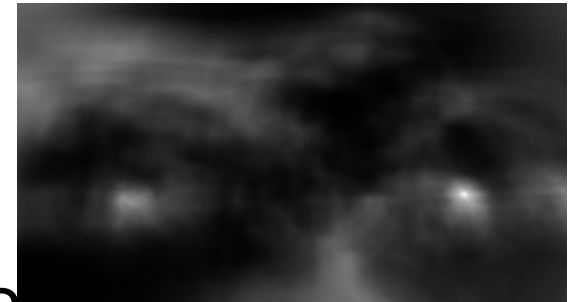
- Limbs attracted to regions of high likelihood (local image evidence is double-counted)



Lan & Huttenlocher, ICCV05

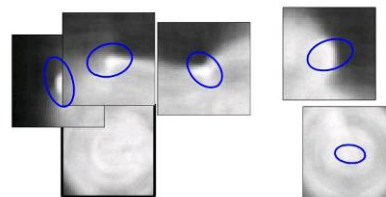
Pictorial Structure Models

- Parts have match quality at each location
 - Location in a configuration space
 - No feature detection

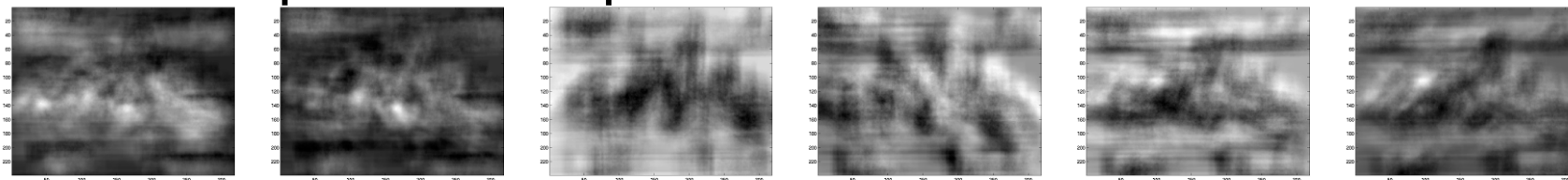


- Maps for parts combined together into overall quality map
 - According to underlying graph structure

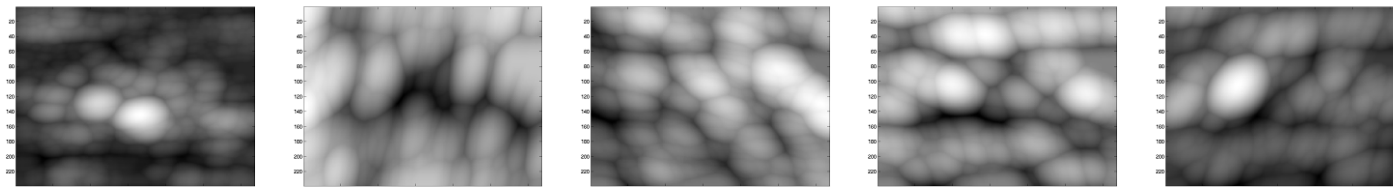
Matching Pictorial Structures



- Cost map for each part



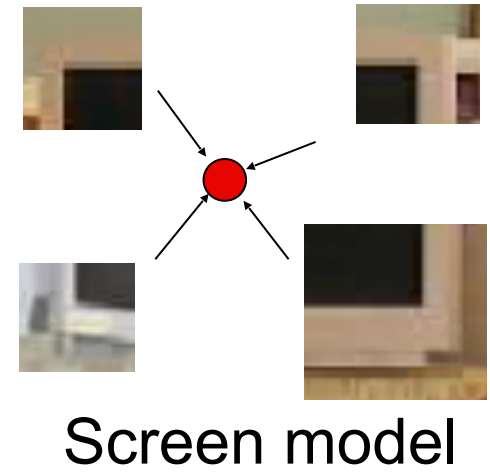
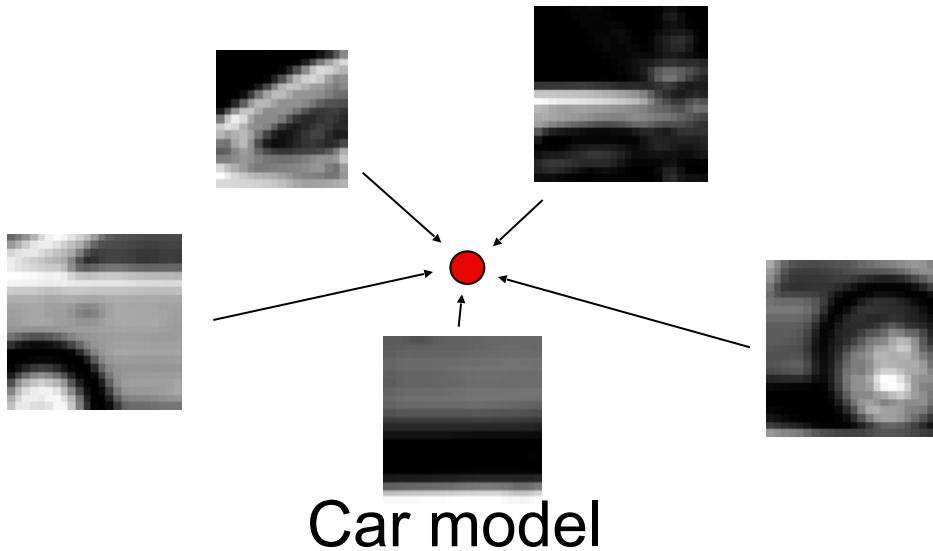
- Distance transform (soft max) using spatial model



- Shift and combine
 - Localize root then recursively other parts

Sparse Part Voting

Part based: We create weak detectors by using parts and voting for the object center location

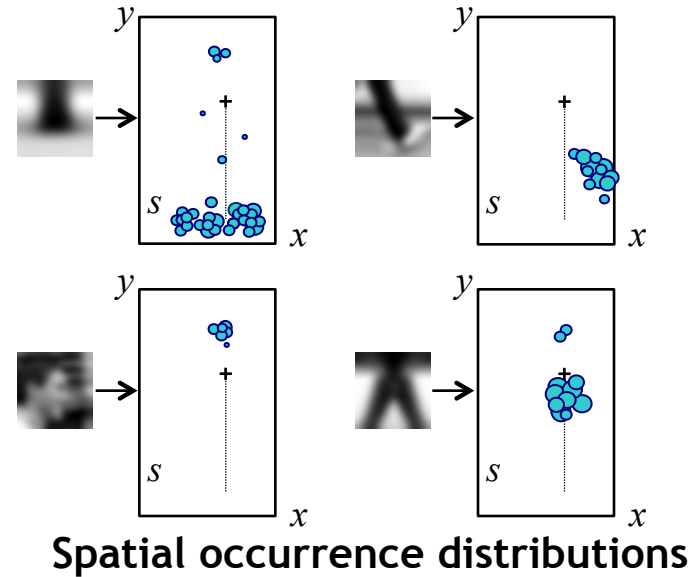
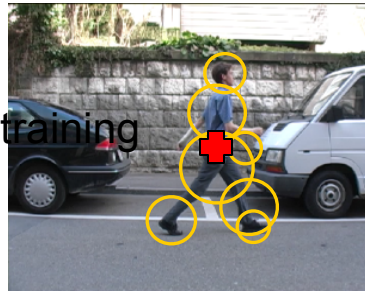


Implicit shape model

- Use Hough space voting to find object
- Leibe and Schiele '03,'05

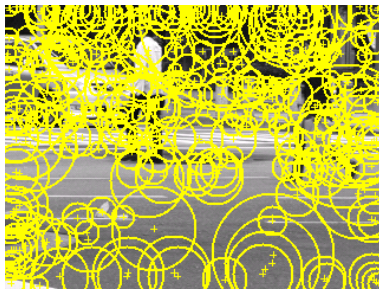
Learning

- Learn appearance codebook
 - Cluster over interest points on training images
- Learn spatial distributions
 - Match codebook to training images
 - Record matching positions on object
 - Centroid is given



Recognition

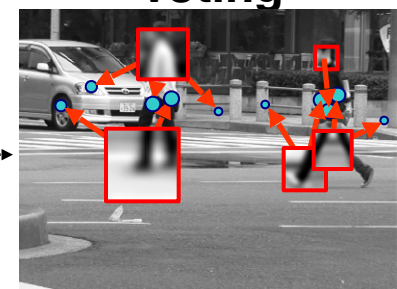
Interest Points



Matched Codebook Entries

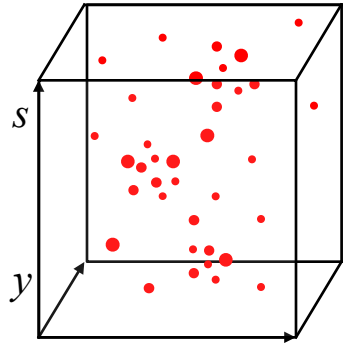


Probabilistic Voting

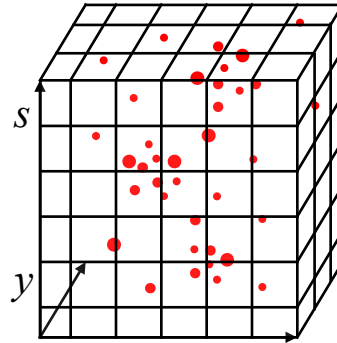


Duality to Sliding Window Approaches...

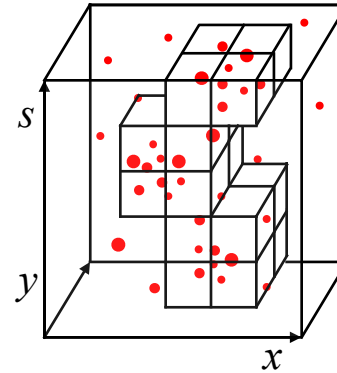
- How to find maxima in the Hough space efficiently?



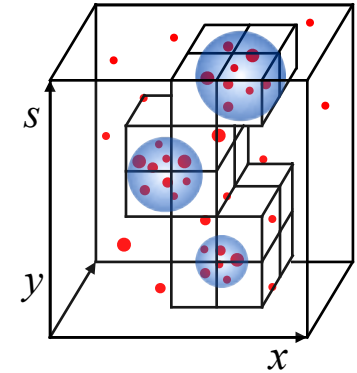
Hough votes



**Binned
accum. array**



**Candidate
maxima**



**Refinement
(MSME)**

- Maxima search = coarse-to-fine sliding window stage!